**Positive charge is distributed uniformly throughout a non-conducting sphere. The highest electric potential occurs:**

- ► **at the center**
- ► at the surface
- ► halfway between the center and surface
- ► just outside the surface

**A total charge of 7×10−8 C is uniformly distributed throughout a non-conducting sphere with a radius of 5 cm. The electric potential at the surface, relative to the potential far away, is about:**

- ► −1.3 × 104 V
- ► **1.3 × 104 V**
- ► 7.0 × 105 V
- ► −6.3 × 104 V

**Eight identical spherical raindrops are each at a potential V , relative to the potential far away. They coalesce to make one spherical raindrop whose potential is:**

- ► V/8
- ► V/2
- ► 2V
- ► **4V**

**A metal sphere carries a charge of 5 × 10−9 C and is at a potential of 400V, relative to the potential far away. The potential at the center of the sphere is:**

- **400V**
- −400V
- $2 \times 10^{-6}$ V
- 0

**A 5-cm radius isolated conducting sphere is charged so its potential is +100V, relative to the potential far away. The charge density on its surface is:**

- $+2.2 \times 10^{-7}$ C/m2
- $-2.2 \times 10^{-7}$ C/m2
- $+3.5 \times 10^{-7}$ C/m2
- **$+1.8 \times 10^{-8}$ C/m2**

**A conducting sphere has charge Q and its electric potential is V , relative to the potential far away. If the charge is doubled to 2Q, the potential is:**

- V
- **2V**
- 4V
- V/4

**The potential difference between the ends of a 2-meter stick that is parallel to a uniform electric field is 400V. The magnitude of the electric field is:**

- zero
- 100V/m
- 200V/m
- **800V/m**

**In a certain region of space the electric potential increases uniformly from east to west and does not vary in any other direction. The electric field:**

- ► points east and varies with position
- ► **points east and does not vary with position**
- ► points west and varies with position
- ► points west and does not vary with position

**If the electric field is in the positive x direction and has a magnitude given by E = Cx2, where C is a constant, then the electric potential is given by V =:**

- ► 2Cx
- ► −2Cx
- ► Cx3/3
- ► **−Cx3/3**

**The work required to carry a particle with a charge of 6.0C from a 5.0-V equipotential surface to a 6.0-V equipotential surface and back again to the 5.0-V surface is:**

- ► **0**
- ► 1.2 × 10−5 J
- ► 3.0 × 10−5 J
- ► 6.0 × 10−5 J

**The equipotential surfaces associated with a charged point particles are:**

- ► radially outward from the particle
- ► vertical planes

- ► horizontal planes
- ► **concentric spheres centered at the particle**

**The electric field in a region around the origin is given by E = C(xˆi + yˆj), where C is a constant. The equipotential surfaces in that region are:**

- ► **concentric cylinders with axes along the z axis**
- ► concentric cylinders with axes along the x axis
- ► concentric spheres centered at the origin
- ► planes parallel to the xy plane

**A particle with charge q is to be brought from far away to a point near an electric dipole. No work is done if the final position of the particle is on:**

- ► the line through the charges of the dipole
- ► **a line that is perpendicular to the dipole moment**
- ► a line that makes an angle of 45◦ with the dipole moment
- ► a line that makes an angle of 30◦ with the dipole moment

**Equipotential surfaces associated with an electric dipole are:**

- ► spheres centered on the dipole
- ► cylinders with axes along the dipole moment
- ► planes perpendicular to the dipole moment
- ► **none of the above**

**The units of capacitance are equivalent to:**

- ► J/C
- ► V/C

► J2/C

► **C2/J**

**A farad is the same as a:**

► J/V

► V/J

► **C/V**

► V/C

**A capacitor C "has a charge Q". The actual charges on its plates are:**

► Q, Q

► Q/2, Q/2

► **Q, −Q**

► Q/2, −Q/2

**Each plate of a capacitor stores a charge of magnitude 1mC when a 100-V potential difference is applied. The capacitance is:**

► 5 μF

► **10 μF**

► 50 μF

► 100 μF

**To charge a 1-F capacitor with 2C requires a potential difference of:**

► **2V**

► 0.2V

► 5V

► 0.5V

**The capacitance of a parallel-plate capacitor with plate area A and plate separation d is given by:**

► 0d/A

► 0d/2A

► **0A/d**

► 0A/2d

**The capacitance of a parallel-plate capacitor is:**

► **proportional to the plate area**

► proportional to the charge stored

► independent of any material inserted between the plates

► proportional to the potential difference of the plates

**The capacitance of a parallel-plate capacitor can be increased by:**

► increasing the charge

► decreasing the charge

► increasing the plate separation

► **decreasing the plate separation**

**If both the plate area and the plate separation of a parallel-plate capacitor are doubled, the capacitance is:**

► doubled

► halved

► **unchanged**

- tripled

**If the plate area of an isolated charged parallel-plate capacitor is doubled:**

- the electric field is doubled
- **the potential difference is halved**
- the charge on each plate is halved
- the surface charge density on each plate is doubled

**If the plate separation of an isolated charged parallel-plate capacitor is doubled:**

- the electric field is doubled
- the potential difference is halved
- the charge on each plate is halved
- **none of the above**

**Pulling the plates of an isolated charged capacitor apart:**

- increases the capacitance
- **increases the potential difference**
- does not affect the potential difference
- decreases the potential difference

**If the charge on a parallel-plate capacitor is doubled:**

- the capacitance is halved
- the capacitance is doubled
- the electric field is halved
- **the electric field is doubled**

**A parallel-plate capacitor has a plate area of 0.2m2 and a plate separation of 0.1mm. To obtain an electric field of 2.0 × 106 V/m between the plates, the magnitude of the charge on each plate should be:**

- ► 8.9 × 10−7 C
- ► 1.8 × 10−6 C
- ► 3.5 × 10−6 C
- ► **7.1 × 10−6 C**

**A parallel-plate capacitor has a plate area of 0.2m2 and a plate separation of 0.1 mm. If the charge on each plate has a magnitude of 4 × 10−6 C the potential difference across the plates is approximately:**

- ► 0
- ► 4 × 10−2 V
- ► 1 × 102 V
- ► **2 × 102 V**

**The capacitance of a spherical capacitor with inner radius a and outer radius b is proportional to:**

- ► a/b
- ► b − a
- ► b2 − a2
- ► **ab/(b − a)**

**The capacitance of a single isolated spherical conductor with radius R is proportional to:**

- **R**
- R2
- 1/R
- 1/R2

**The capacitance of a cylindrical capacitor can be increased by:**

- decreasing both the radius of the inner cylinder and the length
- **increasing both the radius of the inner cylinder and the length**
- increasing the radius of the outer cylindrical shell and decreasing the length
- only by decreasing the length

**A battery is used to charge a series combination of two identical capacitors If the potential difference across the battery terminals is V and total charge Q flows through the battery during the charging process then the charge on the positive plate of each capacitor and the potential difference across each capacitor are:**

- Q/2 and V/2, respectively
- Q and V , respectively
- Q/2 and V , respectively
- **Q and V/2, respectively**

**A battery is used to charge a parallel combination of two identical capacitors. If the potential difference across the battery terminals is V and total charge Q flows through the battery during the charging process then the charge on the**

**positive plate of each capacitor and the potential difference across each capacitor are:**

- ► Q/2 and V/2, respectively
- ► Q and V , respectively
- ► **Q/2 and V , respectively**
- ► Q and V/2, respectively

**A 2-µF and a 1-µF capacitor are connected in series and a potential difference is applied across the combination. The 2-µF capacitor has:**

- ► twice the charge of the 1-µF capacitor
- ► half the charge of the 1-µF capacitor
- ► twice the potential difference of the 1-µF capacitor
- ► **half the potential difference of the 1-µF capacitor**

**A 2-µF and a 1-µF capacitor are connected in parallel and a potential difference is applied across the combination. The 2-µF capacitor has:**

- ► **twice the charge of the 1-µF capacitor**
- ► half the charge of the 1-µF capacitor
- ► twice the potential difference of the 1-µF capacitor
- ► half the potential difference of the 1-µF capacitor

**Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in series must have the same:**

- ► **Q only**
- ► V only

- ► U only
- ► Q and U only

**Let Q denote charge, V denote potential difference, and U denote stored energy. Of these quantities, capacitors in parallel must have the same:**

- ► Q only
- ► **V only**
- ► U only
- ► Q and U only

**Capacitors C1 and C2 are connected in parallel. The equivalent capacitance is given by:**

- ► C1C2/(C1 + C2)
- ► (C1 + C2)/C1C2
- ► C1/C2
- ► **C1 + C2**

**Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by:**

- ► **C1C2/(C1 + C2)**
- ► (C1 + C2)/C1C2
- ► 1/(C1 + C2)
- ► C1/C2

**Capacitors C1 and C2 are connected in series and a potential difference is applied to the combination. If the capacitor that is equivalent to the**

**combination has the same potential difference, then the charge on the equivalent capacitor is the same as:**

- **the charge on C1**
- the sum of the charges on C1 and C2
- the difference of the charges on C1 and C2
- the product of the charges on C1 and C2

## Questions:

1. **An ideal gas is contained in a vessel at 300 K. If the temperature is increased to 900 K, by what factor does each one of the following change?**
   **(a) The average kinetic energy of the molecules.**
   **(b) The rms molecular speed.**
   **(c) The average momentum change of one molecule in a collision with a wall.**
   **(d) The rate of collisions of molecules with walls.**
   **(e) The pressure of the gas.**
2. **Who discover the nucleus? Write the experimental setup that he follows.**
3. **In an analogy between electric current and automobile traffic flow, what would correspond to charge? What would correspond to current?**
4. **(a)When can you expect a body to emit blackbody radiation?**
   **(b) Which law is obeyed by Sun and other stars, briefly explain it.**

5. **Two people are carrying a uniform wooden board that is 3.00 m long and weighs 160 N. If one person applies an upward force equal to 60 N at one end, at what point does the other person lift? Begin with a free-body diagram of the board.**
6. **If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in that region is zero?**
7. **You want to explore the shape of a cel1ain molecule by scattering electrons of momentum p from a gas of the molecules and studying the deflection of the electrons.**
8. **You will be able to see finer details in the molecules by (a) increasing p; (b) decreasing p: (c) not worrying what p is.**
9. **A vessel is filled with gas at some equilibrium pressure and temperature. Can all gas molecules in the vessel have the same speed?**
10. **What are the properties of wave function?**